

Software Automation



Property Price Prediction Model

Part C - Discussion

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Impact of the model on the individual, society and environment

Impact on the individual

The Property Price Prediction model is targeted towards individuals (specifically first home buyers), meaning it will have a significant impact on them and their decisions. Being able to access a model trained on real property data allows buyers to use the model in order to gain insight on whether the listed / guide price for a property is reasonable given its proximity to the CBD. This is useful as it allows buyers to not rely entirely on the word of real estate agents which may not always have the buyer's best interest at heart. However, if the model is inaccurate about a specific property due to it being an outlier, it could negatively impact the individual as it could lead them to potentially avoid further looking into a property predicted to have a higher selling price than it actually did and vice versa.

Though not the target market for the model, sellers and workers in real estate can also benefit from the model, as it would allow sellers to form a realistic listing / guide price, and assist real estate workers with processes like market analysis.

Impact on society

The model's potentially large impact on society means it will have implications on the economy and distribution of wealth. A property price predicting model like this one could put ordinary home buyers on the same level as domestic and foreign investors as the model can bridge the gap between the pricing information investors may have compared to the lack of information a first time home buyer may have.

In addition to this, the model can also be extremely beneficial to people with disabilities. Having a model which can provide an estimated price on a property in relation to its proximity to the CBD is valuable to people with disabilities since the closer the location is to the city centre, the greater the access to transport, medical and other services is, facilities that are harder to access the further away someone may live. Being able to then put a price on the property allows these people to choose a suitable location within their budget.

Impact on the environment

The model's direct environmental impact is limited due to the scale the model was created on. The processes leading up to and including the model's deployment such as data wrangling, model training, etc. has a small and limited carbon footprint to that of large language models. The direct impact of the model would increase significantly if it reached a similar scale to large language models due to the amount of energy consumption of the data centres the model runs on.

The model could also have indirect impacts on the environment in addition to these. The model's pricing predictions could influence urban development around what the model predicts to be higher priced areas, causing increase in greenhouse gas emissions, habitat loss and other related environmental impacts.

How patterns in human behaviour influence the model

Psychological responses

Human behaviour has a significant influence on the performance of the Property Price Predictor model, since human behaviour in the property market is often influenced by psychological factors which are not always captured in data. Examples of this include the concept of herd mentality. In the property market, this can mean that buyers are motivated to buy property during periods of booms, leading to the prices paid for housing exceeding the model's predictions. The training data used to train the model could have potentially picked up these booms (or recessions depending on the trends in the period the data was recorded) and begin applying these to its predictions. This influences the model as it effects the model's reliability due to the fact that it will make predictions based on these patterns in human behaviour leading it to overpredict or underpredict due to the fit of the polynomial regression line.

Patterns related to acute stress response

Another human behaviour influencing the model is acute stress conditions. In the property market this would most likely relate to financial hardship, causing sellers to accept low prices for high value properties to increase the speed of getting the property sold to secure funds. These types of property listings create outliers in data, which in relation the polynomial regression model the price predictor is, can cause the prediction line to lean towards these values if a large amount accumulate.

Cultural protocols & Belief systems

In addition to these, cultural protocols also heavily influence the model, as different belief systems can drive prices upwards such as proximity to churches, mosques, or places carrying cultural significance to groups of people, factors which are hard to pick up on with regular features such as the ones the model is trained on. People who have deeply held cultural protocols and belief systems may not care about proximity to the CBD, rather proximity to their respective sites.

Bias in relation to the model

Human bias

As previously discussed, human behaviour can influence the model causing bias in its predictions. In addition to the examples previously provided, there are other ways humans bias machine learning models like the property price predictor due to their behaviour. This includes willingness to pay more for certain properties due to other features excluding cultural protocols and belief systems or even distance from the CBD. This includes proximity to bare necessities like grocery stores, or the preference of having many parking spaces. These are all factors which create significant outliers in the data causing the model to favour and lean towards them.

Dataset source bias

However, human behaviour is not the only significant effect on the model when it comes to bias, since dataset source bias can include inherent biases before development even begins. In the context of the property price prediction model, the dataset used to train the model was sourced from domain.com.au, and in a specific timeframe. This means that the properties the model is trained on are only properties listed on Domain, excluding things like off market sales and even other website data. Off market sales are also extremely common for high value properties where the buyer and seller choose to avoid publicity as much as possible, meaning that the high end of prices are underrepresented in the dataset.

Model bias

Model bias is also a key factor. The property price predictor is a polynomial regression model fitted with a degree of 3. The non linear prediction line has an advantage over a typical linear regression model as it doesn't assume a linear relationship in the data. However, polynomial regression introduces the risk of overfitting and underfitting. Since the model has a degree of 3, it sits in between a good fit and underfitting. The reason for this is because a degree of 4-5 or higher starts overfitting, causing the model to extrapolate prices based on a few outliers, and a degree of 2 performs too similarly to a linear regression model leading to a more significant underfitting. However, due to the model's poor performance, we see that a degree of 3, while being the best fit for a polynomial regression model, still performs poorly on both training and testing data. This means that the model has low variance, signalling that it is stable, though too simple for the complexity of the dataset.

In future development, these biases can be addressed and mitigated by changing the model from a polynomial regression algorithm to a KNN (K-Nearest Neighbour) algorithm, working by calculating the distance between a new data point and existing points, selecting the closest "neighbours" to determine the result. This would be useful in a dataset like this in order to determine what the price of a property may be based on surrounding prices of properties in its neighbourhood. In addition to this, the limitations of the model should also be displayed on the website to ensure that users know that predictions the model makes may not be entirely accurate and to consider this before making big financial decisions.